

NIDS Session Report

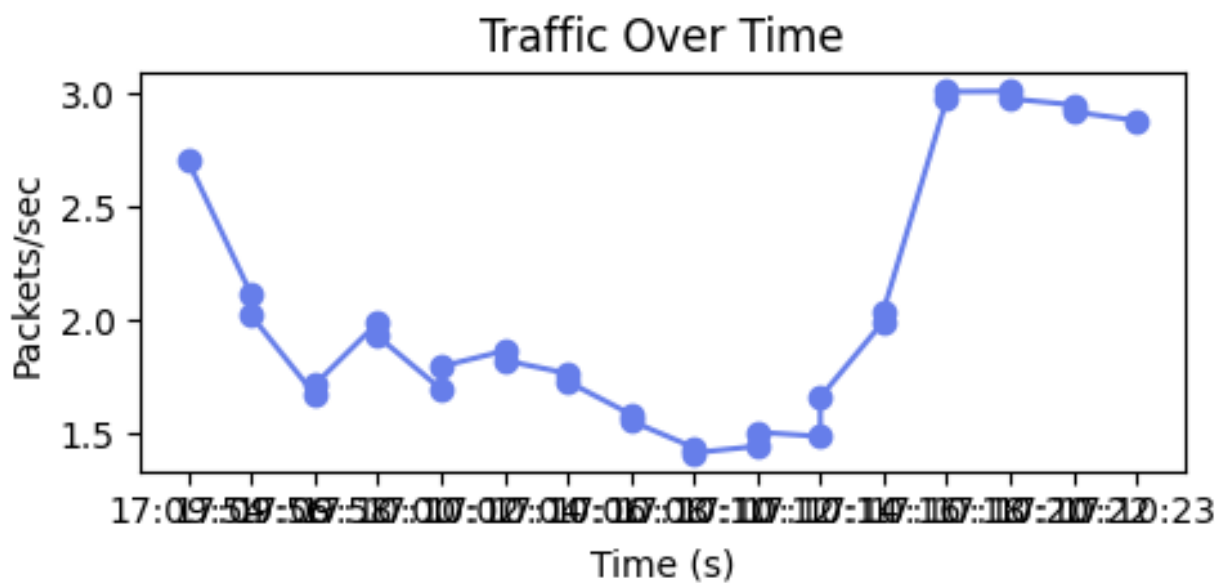
AI-Based Network Intrusion Detection System

This report summarizes your network activity and any suspicious events detected. It's designed to be easy to understand, even if you're not a network expert.

How to Read This Report

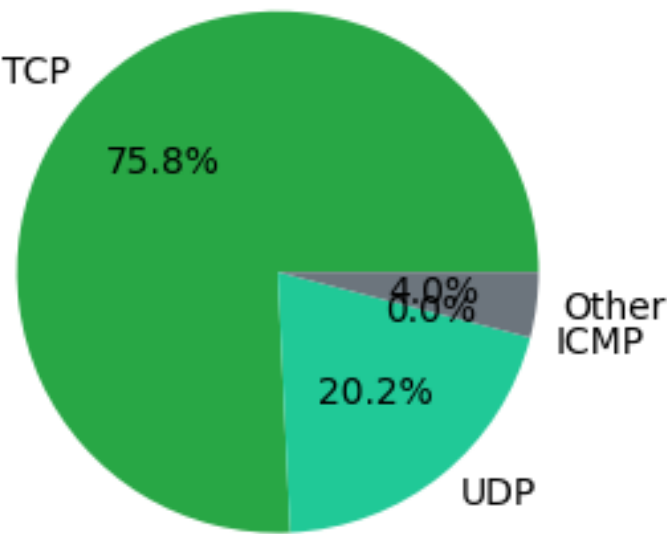
- Packets: Each piece of data sent over your network.
- Anomaly: Something unusual or unexpected in the network traffic.
- Alert: A warning about a specific suspicious activity.

Traffic Over Time



Protocol Distribution

Protocol Distribution



Session Summary

Start Time:	1750246788.83528
Stop Time:	2025-06-18T17:10:23.199966
Uptime:	34 seconds
Packets Processed:	99
Anomalies Detected:	6
Packet Rate:	2.88 packets/sec

Protocol Statistics

TCP PACKETS:	75
UDP PACKETS:	20
ICMP PACKETS:	0
OTHER PACKETS:	4

Security Alerts (3)

Type: Port Scan | Time: 2025-06-18T17:09:53.057240

Details: {'src_ip': '10.0.2.15', 'dst_ip': '142.250.195.196', 'dst_port': 443, 'protocol': 17}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: MI Anomaly | Time: 2025-06-18T17:09:53.874565

Details: {'anomaly_score': -3.552933588544145, 'features': {'packet_count': 16, 'total_bytes': 9900, 'avg_packet_length': np.float64(618.75), 'std_packet_length': np.float64(563.7026587661264), 'min_packet_length': 66, 'max_packet_length': 1292, 'avg_payload_size': np.float64(576.75), 'std_payload_size': np.float64(563.7026587661264), 'total_payload_size': 9228, 'tcp_count': 0, 'udp_count': 16, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 1, 'tcp_ratio': 0.0, 'udp_ratio': 1.0, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 2, 'unique_dst_ports': 2, 'total_ports_accessed': 2, 'common_port_ratio': 0.4375, 'high_port_ratio': 0.5625, 'avg_dst_port': np.float64(30104.75), 'std_dst_port': np.float64(26159.204650323372), 'min_dst_port': 443, 'max_dst_port': 53175, 'unique_connections': 2, 'total_connection_attempts': 16, 'avg_packets_per_connection': 8.0, 'connection_diversity': 0.125, 'max_packets_per_connection': 9, 'unique_src_ips': 2, 'unique_dst_ips': 2, 'src_ip_diversity': 0.125, 'dst_ip_diversity': 0.125, 'private_src_ip_ratio': 0.5, 'private_dst_ip_ratio': 0.5, 'packet_rate': 16.0, 'inter_arrival_time_mean': np.float64(0.004876152674357096), 'inter_arrival_time_std': np.float64(0.007825480302994535), 'burst_count': 10, 'burst_ratio': 0.6666666666666666, 'payload_size_mean': np.float64(576.75), 'payload_size_std': np.float64(563.7026587661264), 'zero_payload_ratio': 0.0, 'large_payload_ratio': 0.375, 'payload_size_entropy': np.float64(1.5306390622295665), 'syn_count': 0, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 0, 'urg_count': 0, 'syn_ratio': 0, 'fin_ratio': 0, 'rst_ratio': 0}, 'timestamp': '2025-06-18T17:09:53.874550', 'packet_count': 16, 'protocol_distribution': {'tcp': 0, 'udp': 16, 'icmp': 0}, 'connection_info': {'unique_connections': 2, 'unique_src_ips': 2, 'unique_dst_ips': 2}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

Type: Ddos Indicator | Time: 2025-06-18T17:10:17.161663

Details: {'src_ip': '3.229.150.19', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: A large number of packets were sent to a single device, possibly overwhelming it. This is a common sign of a Distributed Denial of Service (DDoS) attack, which can take down services.

What Should I Do?

- If you see many alerts: Consider checking your devices for unusual activity or malware.
- If you see DDoS or Port Scan alerts: Make sure your firewall is active and up-to-date.
- If unsure: Share this report with your IT support or a cybersecurity professional.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
Port Scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.

DDoS Indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
Unusual Protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ML Anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.